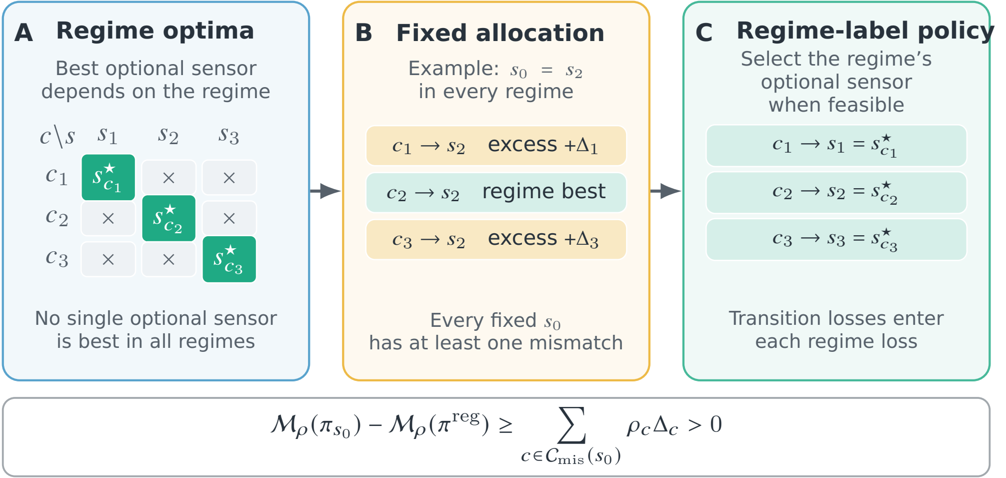
A Regime optima
Best optional sensor depends on the regime
$c \backslash s$ s_1 s_2 s_3
c_1 $s^\star_{c_1}$ × ×
c_2 × $s^\star_{c_2}$ ×
c_3 × × $s^\star_{c_3}$
No single optional sensor is best in all regimes
B Fixed allocation
Example: $s_0 = s_2$ in every regime
$c_1 \to s_2$ excess $+\Delta_1$
$c_2 \to s_2$ regime best
$c_3 \to s_2$ excess $+\Delta_3$
Every fixed s_0 has at least one mismatch
C Regime-label policy
Select the regime’s optional sensor when feasible
$c_1 \to s_1 = s^\star_{c_1}$
$c_2 \to s_2 = s^\star_{c_2}$
$c_3 \to s_3 = s^\star_{c_3}$
Transition losses enter each regime loss
$\mathcal{M}_\rho(\pi_{s_0}) - \mathcal{M}_\rho(\pi^{\mathrm{reg}}) \geq \sum_{c \in \mathcal{C}_{\mathrm{mis}}(s_0)} \rho_c \Delta_c > 0$